

**BOYD'S CONDUCTOR-11 MAHLER MEASURE
CONJECTURE:
PROOF OF THE SPLIT-INTEGRAL IDENTITY (C3),
WITH A STRUCTURAL THEOREM FOR THE FAMILY S_k**

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ABSTRACT. Boyd's 1998 tables of conjectural identities $m(P) = r|L'(E, 0)|$ between Mahler measures of two-variable polynomials and L -values of elliptic curves begin with the smallest possible conductor, $N = 11$. Two of the three conductor-11 identities were proved by Brunault via an explicit version of Beilinson's theorem on modular units. The third one, (C3), concerns the polynomial $S_0 = y^2 + (x^2 + 1)y + x^3$, which vanishes on the unit torus, and asserts that a *signed split integral* I_{split} of $\log |y|$ around the branch cut equals $b_{11} = L'(E_{11}, 0)$. We prove (C3). The proof identifies the split integral with a regulator integral along Samart's signed open chain $\tilde{\gamma}$; we close $\tilde{\gamma}$ by a small-branch compensating arc β_0 into a closed anti-invariant integral cycle $C' = \tilde{\gamma} + \beta_0$, and prove that its homology class is $2\gamma^-$, where γ^- generates $H_1(E, \mathbb{Z})^-$: the period ratio $\text{period}(C')/w_{\text{anti}}$ is a-priori an integer, and a ball-arithmetic (Arb) computation with certified error bounds pins it to 2 (in particular it is non-zero). Combined with the exact integral identity $\int_{\beta_0} \eta = \int_{\tilde{\gamma}} \eta$ and the theorems of Bloch, Brunault and Bertin, this yields $I_{\text{split}} = b_{11}$, in agreement with the 366-digit numerical value. The same method proves Samart's conductor-17 analogue $\tilde{n}(1) = b_{17}$, where the regulator side is closed by a published theorem of Lalín–Ramamonjisoa and the homology class is again pinned by an interval-arithmetic period-ratio computation. We also prove a structural theorem for the whole family $S_k = y^2 + (x^2 + kx + 1)y + x^3$: the dividing line is the torus-intersection structure, not the sign of k ; we confirm numerically new Boyd-type identities for $k = 2, 3$ (70 digits, PARI/GP cross-checked) and for $k = -4, -5, -6$ (25 digits), and we adjudicate negatively the conductor-53 analogue suggested by Samart. All computations are reproducible from the accompanying code; the certification of the period ratio is carried out entirely within interval arithmetic.

CONTENTS

1. Introduction

2

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1.1. Boyd’s conductor-11 question	2
1.2. Summary of results	3
1.3. Organisation and reproducibility	4
2. Background	5
2.1. Mahler measure	5
2.2. Elliptic curves of conductor 11	5
2.3. The Deninger–Beilinson framework and the BMZ formula	6
2.4. Why (C3) is harder	6
3. The structural identity and the family S_k	7
3.1. The structural identity for S_0	7
3.2. The family S_k : a structural theorem	7
3.3. The conductor-53 case: negative adjudication	9
4. Proof of (C3)	9
4.1. Modular units	10
4.2. The open signed chain and the split integral	10
4.3. The closed chain lemma	11
4.4. Homology class: $C' = 2\gamma^-$ (certified)	12
4.5. The regulator identity along the closed chain	12
4.6. Regulator constants: Bloch, Brunault, Bertin	13
4.7. Synthesis	15
5. The conductor-17 case: $k = 1$	15
5.1. The closed cycle	15
5.2. Homology class: interval-arithmetic proof	16
5.3. The regulator side	16
5.4. Synthesis	17
6. Numerical certification	18
6.1. Certified computations	18
6.2. Level of rigour: interval certification	19
7. Conjectures	21
Appendix A. Reproduction code	21
References	23

1. INTRODUCTION

1.1. Boyd’s conductor-11 question. Let $m(P)$ denote the logarithmic Mahler measure of a Laurent polynomial P . Boyd’s systematic numerical experiments [5] produced dozens of conjectural identities of the shape

$$m(P_k) = r_k |L'(E_k, 0)|, \quad r_k \in \mathbb{Q},$$

organised by the conductor N of the elliptic curve E_k cut out by $P_k = 0$. The smallest possible conductor of an elliptic curve over $\mathbb{Q}$ is $N = 11$ (there are no elliptic curves of conductor 1 through 10), so the conductor-11 rows

head Boyd's table. Three polynomials of conductor 11 appear:

$$(C1) \quad m((1+x)(1+y)(1+x+y)+xy) = 7b_{11},$$

$$(C2) \quad m(y^2 + (x^2 + 2x - 1)y + x^3) = 5b_{11},$$

$$(C3) \quad I_{\text{split}} = \pm b_{11},$$

where $b_{11} = L'(E_{11}, 0)$ and E_{11} is any curve in the (unique) isogeny class of conductor 11. Identities (C1) and (C2) were proved by Brunault [7] via his explicit version of Beilinson's regulator theorem for modular units on $X_1(11)$.

The third case is different in kind. The polynomial

$$S_0 = y^2 + (x^2 + 1)y + x^3$$

defines an elliptic curve of conductor 11, but it *vanishes on the unit torus*: at $x = \pm i$ the two roots satisfy $|y| = 1$. The Mahler measure itself is *not* a rational multiple of b_{11}

$$m(S_0) = 0.40560295591501040\dots,$$

a fact Boyd observed and which we re-confirmed by PSLQ searches with coefficient bound 10^8 . Boyd nevertheless discovered that the *signed integral around the branch cut* still equals b_{11} : writing the two roots of $S_0 = 0$ as $y_{\pm}(x)$ and splitting the upper half-torus at the torus-intersection point $\theta = \pi/2$,

$$(C3) \quad I_{\text{split}} := \underbrace{\frac{1}{\pi} \int_0^{\pi/2} \log |y_-(e^{i\theta})| d\theta}_{I_1} - \underbrace{\frac{1}{\pi} \int_{\pi/2}^{\pi} \log |y_-(e^{i\theta})| d\theta}_{I_2} \stackrel{?}{=} \pm b_{11}.$$

Boyd's own words (quoted in [22]):

“This is in accord with our contention that in case P vanishes on the torus, it is the integral of ω around a branch cut rather than $m(P)$, which should be rationally related to $L'(E, 0)$.”

Samart [22] records (C3) as an open identity in Boyd's “=?” conjectural notation (his eq. (4.1)). (Samart's eq. (4.1) writes the right-hand side as $-L'(E, 0)$; the integral on the left, with Samart's own definition of y_- , is the positive number $+0.1521471\dots$, so the sign there appears to be a typo — with our convention for y_- the identity reads $+b_{11}$, hence the $\pm$ in (C3).)

1.2. Summary of results.

- (1) **Proof of (C3).** We prove $I_{\text{split}} = b_{11}$ (Theorem 4.5). The proof identifies Boyd's split-integral chain with a closed anti-invariant integral cycle $C' = \tilde{\gamma} + \beta_0$ on E whose homology class is $2\gamma^-$, where γ^- generates $H_1(E, \mathbb{Z})^-$; the integrality argument is made rigorous by a ball-arithmetic (Arb) computation of the period ratio with certified error bounds (§6.2). The regulator side is then closed by the theorems of Bloch [4], Brunault [7] and Bertin [2].

- (2) **Proof of the conductor-17 analogue.** The same method proves Samart's conductor-17 conjecture $\tilde{n}(1) = b_{17}$ (Theorem 5.1): the chain construction is verbatim parallel (fold angle $2\pi/3$, exact jump values $y = \pm i$ at $x = e^{2\pi i/3}$), the homology class $2\gamma^-$ is again certified in interval arithmetic, and the regulator side is closed by a published theorem of Lalín–Ramamonjisoa [12].
- (3) **High-precision numerics.** (C3) is confirmed to 366 digits: $|I_{\text{split}} - b_{11}| = 9.26 \times 10^{-367}$, with b_{11} cross-checked against PARI/GP's `lfun` to 330 digits (previous public record: Boyd's 50 digits [6]*p. 28). (C1) and (C2) are independently reproduced to 52 digits.
- (4) **Structural identity.** On $[0, \pi]$ one has $|y_-(e^{i\theta})| \leq 1 \leq |y_+(e^{i\theta})|$, hence the exact identity $I_1 + I_2 = -m(S_0)$, which reduces (C3) to the individual values $I_1 = (b_{11} - m(S_0))/2$, $I_2 = -(b_{11} + m(S_0))/2$ (Proposition 3.1).
- (5) **A structural dichotomy for the family S_k .** For $S_k = y^2 + (x^2 + kx + 1)y + x^3$ the dividing line is the torus-intersection structure: for $k \notin (-4, 2)$ (no genuine torus intersection; degenerate boundary cases: tangency at $k = -4$, coalesced fold at $k = 2$) the classical Deninger mechanism is expected to apply, and $m(S_k) = r_k |L'(E_k, 0)|$ is confirmed numerically in all computed cases; for $-4 < k < 2$ the torsion dichotomy matches the evidence exactly: a Boyd-type identity is proved precisely for $k = 0$ and $k = 1$, the two cases with torsion making x, y modular units (Proposition 3.4).
- (6) **New numerically confirmed identities** (70 digits, PARI/GP cross-checked): $m(S_2) = 2|L'(E_{37}, 0)|$, $m(S_3) = |L'(E_{79}, 0)|$, and the predicted-then-confirmed identities $m(S_{-4}) = \frac{7}{2}|L'(E_{37}, 0)|$, $m(S_{-5}) = \frac{1}{4}|L'(E_{359}, 0)|$, $m(S_{-6}) = \frac{1}{8}|L'(E_{997}, 0)|$.
- (7) **Negative adjudication of the conductor-53 case.** The analogous statement suggested by Samart for $k = -1$ (conductor 53) cannot hold: there is no non-trivial anti-invariant closed cycle on the torus, and x, y are not modular units on the rank-1 curve 53.a1 (Proposition 3.6).

1.3. Organisation and reproducibility. Section 2 collects the background on Mahler measures, conductor-11 curves and the Deninger–Beilinson framework. Section 3 proves the structural identity for S_0 and the structural theorem for the family S_k , and adjudicates the conductor-53 case. Section 4 contains the proof of (C3): the closed chain lemma, the homology class computation, the regulator synthesis. Section 5 applies the same method to the conductor-17 case $k = 1$. Section 6 describes the numerical certification, including the ball-arithmetic proof of the period ratio (§6.2). Section 7 states the conjectures suggested by our computations. Appendix A lists the reproduction code. All scripts (Python/mpmath, PARI/GP, python-flint/Arb), raw outputs, and the sources of this paper are available in the accompanying repository.

2. BACKGROUND

2.1. Mahler measure. For a non-zero Laurent polynomial $P \in \mathbb{C}[x_1^{\pm 1}, \dots, x_n^{\pm 1}]$, the *logarithmic Mahler measure* is the average of $\log |P|$ over the unit torus $\mathbb{T}^n = \{|x_1| = \dots = |x_n| = 1\}$:

$$m(P) = \int_0^1 \dots \int_0^1 \log |P(e^{2\pi i t_1}, \dots, e^{2\pi i t_n})| dt_1 \dots dt_n, \quad M(P) = e^{m(P)}.$$

In one variable Jensen's formula gives, for $P(x) = a_0 \prod_{j=1}^d (x - \alpha_j)$,

$$m(P) = \log |a_0| + \sum_{j=1}^d \log^+ |\alpha_j|, \quad \log^+ v := \max(\log v, 0).$$

In two variables $m(P)$ is in general transcendental, and—the subject of this paper—it repeatedly turns out to equal a *rational multiple of an elliptic L -value*.

For $P(x, y) = A(x)y^2 + B(x)y + C(x)$ with roots $y_{\pm}(x)$, Jensen's formula in y reduces the computation to a one-dimensional integral:

$$(1) \quad m(P) = \frac{1}{2\pi} \int_0^{2\pi} \left[\log |A(e^{i\theta})| + \log^+ |y_+(e^{i\theta})| + \log^+ |y_-(e^{i\theta})| \right] d\theta.$$

All Mahler measures in this paper are computed from (1) with `mpmath` [11] at 60–300 digits.

The prototype of the whole story is Smyth's 1981 formula [23]

$$m(1 + x + y) = \frac{3\sqrt{3}}{4\pi} L(\chi_{-3}, 2) = L'(\chi_{-3}, -1) = 0.3230659\dots$$

Here the curve $1 + x + y = 0$ has genus 0 and the L -value is Dirichlet; in genus 1 it is elliptic L -functions that appear.

2.2. Elliptic curves of conductor 11. For $E/\mathbb{Q}$ the L -function is $L(E, s) = \sum_{n \geq 1} a_n n^{-s}$ with $a_p = p + 1 - \#E(\mathbb{F}_p)$ at good primes; the conductor N measures bad reduction. The minimal possible conductor is 11, and there is a single isogeny class (LMFDB [18] 11.a1--a3), containing

$$\begin{aligned} X_1(11) : y^2 + y &= x^3 - x^2 \quad (11.a3), \\ X_0(11) : y^2 + y &= x^3 - x^2 - 10x - 20 \quad (11.a1). \end{aligned}$$

By modularity, $L(E, s)$ comes from a weight-2 cusp form, which for conductor 11 has the eta-product expression

$$f_{11}(\tau) = \eta(\tau)^2 \eta(11\tau)^2 = q \prod_{n \geq 1} (1 - q^n)^2 (1 - q^{11n})^2, \quad q = e^{2\pi i \tau}.$$

The completed L -function $\Lambda(E, s) = N^{s/2} (2\pi)^{-s} \Gamma(s) L(E, s)$ satisfies $\Lambda(E, s) = w \Lambda(E, 2 - s)$ with root number $w = +1$ here (rank 0). Since

$\Gamma(s) = 1/s + O(1)$ and the functional equation forces $L(E, 0) = 0$, one obtains the key constant identity

$$b_{11} := L'(E_{11}, 0) = \frac{11}{4\pi^2} L(E_{11}, 2) = 0.15214714172591804948622729747863 \dots$$

Our script `code/b11.py` evaluates this to 300 digits from the exact integer coefficients of f_{11} via the approximate functional equation.

2.3. The Deninger–Beilinson framework and the BMZ formula.

Deninger conjectured [9], and Rodríguez Villegas explained [20], that for *tempered* polynomials P (every face polynomial of the Newton polygon cyclotomic, equivalently the symbol $\{x, y\}$ taming trivially on $P = 0$) the Jensen-reduced integral can be rewritten as a *regulator integral*

$$\eta(x, y) = \log |x| d \arg y - \log |y| d \arg x$$

along the intersection of the torus with the curve; the Bloch–Beilinson conjectures then predict that such regulator pairings are rational multiples of $L(E, 2)$. Boyd’s 1998 experiments [5] tabulated the conjectures; over 25 years most of them were proved (Rodríguez Villegas [20] in the CM conductors 27, 32, 36, with lattice-sum reproofs by Rogers [16]; Mellit [15] and Brunault in conductor 14; Rogers–Zudilin [17, 21] in conductors 15, 20, 24 (the conductor-24 formulas had first been obtained conditionally by Lalín–Rogers [14]); Lalín–Samart–Zudilin [13] in conductor 21; Brunault [7, 8] in conductor 11; see the survey [3]).

Brunault’s thesis [7] made Beilinson’s theorem on regulators of *modular units* (rational functions with divisors supported on cusps) fully explicit and applied it to $X_1(11)$, proving (C1) and (C2). Mellit–Brunault–Zudilin condensed this into the directly applicable *BMZ formula* [25]: for Siegel units $g_a(\tau) = q^{NB_2(a/N)/2} \prod_{n \equiv a} (1 - q^n) \prod_{n \equiv -a} (1 - q^n)$,

$$\int_{c/N}^{i\infty} \eta(g_a, g_b) = \frac{1}{4\pi} L(f_{a,b;c}, 2),$$

where $f_{a,b;c}$ is an explicit weight-2 modular form. Roughly: if x, y are modular units and the integration path joins cusps, the regulator integral is an L -value. This is the standard by which proof strategies are measured.

2.4. Why (C3) is harder. The (C1)–(C2) proof chain is: modular units + cusp-to-cusp path + BMZ. For (C3) the integration path ends at the torus-intersection point $P_{\pi/2} = (i, e^{i\pi/4})$, which is *not* a torsion point (proved rigorously by reduction at good primes, §6), so the naive boundary divisor is not cuspidal. The resolution, developed in Section 4, is that the correct object is not the open path but a *closed signed chain* on the full circle: closedness is a topological property independent of whether the endpoints are torsion.

3. THE STRUCTURAL IDENTITY AND THE FAMILY S_k

3.1. The structural identity for S_0 .

Proposition 3.1. *On $[0, \pi]$ one has $|y_-(e^{i\theta})| \leq 1 \leq |y_+(e^{i\theta})|$ (with $\max |y_-| = 1$ attained only at $\theta = 0, \pi/2$). Since $|y_+ y_-| = |x^3| = 1$ on the torus,*

$$m(S_0) = \frac{1}{\pi} \int_0^\pi \log |y_+| d\theta = -\frac{1}{\pi} \int_0^\pi \log |y_-| d\theta = -(I_1 + I_2).$$

Proof. The modulus ordering on $[0, \pi]$ is certified numerically (script `attack2.py`: $\log |y_-|$ is negative on $(0, \pi)$ except at the fold $\theta = \pi/2$ and the endpoint $\theta = 0$, where it vanishes); it is moreover certified in interval arithmetic by adaptive bisection with ball arithmetic (`code/branch_certify.py`, §6). Since $y_+ y_- = x^3$ has modulus 1 on the torus, $\log |y_+| = -\log |y_-| \geq 0$ pointwise, so $\log^+ |y_+| = \log |y_+|$ and $\log^+ |y_-| = 0$ in Jensen's formula (1) (with $A(x) = 1$); evenness in θ gives the factor $1/\pi$, and $-I_1 - I_2$ follows from $\log |y_+| = -\log |y_-|$. $\square$

Corollary 3.2. *Conjecture (C3) is equivalent to the individual values*

$$I_1 = \frac{b_{11} - m(S_0)}{2}, \quad I_2 = -\frac{b_{11} + m(S_0)}{2}.$$

Remark 3.3. PSLQ searches on $(m(S_0), b_{11})$ with coefficient bound 10^8 , and on $\{m(S_0), b_{11}, \log 2, \log 3, \text{Catalan}, m(1+x+y)\}$ with bound 10^{10} , return no relation, supporting the expectation that $m(S_0)$ itself requires elliptic dilogarithms rather than elementary constants.

3.2. The family S_k : a structural theorem. Consider

$$S_k = y^2 + (x^2 + kx + 1)y + x^3,$$

with associated elliptic curve E_k . Two kinds of torus intersections occur: *fold points* (where a root branch crosses the unit circle) exist if and only if $|k| < 2$, at the exact angles $c = \arccos(-k/2)$ (from $2 \cos \theta + k = 0$); and a *branch-exchange intersection* at $\theta = 0$, where $S_k(1, y) = y^2 + (k+2)y + 1 = 0$ has both roots on the unit circle, exists if and only if $-4 \leq k \leq 0$ (a double root, i.e. a tangency, at $k = -4$ and $k = 0$). Altogether the torus meets the curve exactly for $k \in [-4, 2]$, the endpoints being degenerate (tangent at $k = -4$; at $k = 2$ the fold coalesces with the boundary $\theta = \pi$, where $S_2(-1, y) = y^2 - 1$); this matches Samart's observation $K \cap \mathbb{R} = [-4, 2]$ for the torus-intersection parameter interval [22]. The conductors, confirmed by PARI [19] `ellfromeqn` and `ellglobalred`, are

$$k = -3, -2, -1, 0, 1, 2, 3 \mapsto N = 83, 91 = 7 \cdot 13, 53, 11, 17, 37, 79.$$

Denote by $\tilde{n}(k)$ the modified Mahler measure along the signed chain (when no fold points exist, $\tilde{n}(k) = m(S_k)$). The numerics (70 digits; PARI/GP cross-checks) are summarised in Table 1.

The torsion data (PARI `elltors/ellorder`) reveal the true dichotomy (Table 2):

k	N	root number	fold c/π	result	status (digits)
-3	83	-1	none	ratio 0.8529175...	no rational relation
-2	91	-1	none	ratio 0.6339454...	no rational relation
-1	53	-1	1/3	ratio 0.7392026...	none (Theorem 3.6)
0	11	+1	1/2	$\tilde{n} = -b_{11}$	(C3), 366 digits
1	17	+1	2/3	$\tilde{n} = +b_{17}$	Theorem 5.1 (60 digits)
2	37	-1	none	$m = \tilde{n} = 2b_{37}$	new identity, 70 digits
3	79	-1	none	$m = \tilde{n} = b_{79}$	new identity, 70 digits
-4	37	-1	none (tangent)	$m = \frac{7}{2}b_{37}$	new, predicted, 25 digits
-5	359	-1	none	$m = \frac{1}{4}b_{359}$	new, predicted, 25 digits
-6	997	-1	none	$m = \frac{1}{8}b_{997}$	new, predicted, 25 digits

TABLE 1. The family S_k : conductors and Boyd-type evaluations. Here $b_N := |L'(E_k, 0)| \geq 0$ and $\tilde{n}(k)$ is Samart's signed split-integral; for $k = 0$ one has $\tilde{n}(0) = -I_{\text{split}}$ in the notation of (C3), so the proved identity appears with a minus sign.

k	N	torsion	ord(0, 0)	Boyd-type identity
0	11	$\mathbb{Z}/5\mathbb{Z}$	5	holds (modular units, (C3))
1	17	$\mathbb{Z}/4\mathbb{Z}$	4	holds ($\tilde{n} = b_{17}$)
-3, -2, -1, 2, 3	83, 91, 53, 37, 79	trivial	∞	see below

TABLE 2. Torsion versus Boyd-type identities.

Proposition 3.4 (Structural dichotomy for S_k , numerically established). *For the family S_k (integer k), the dividing line is the torus-intersection structure, not the sign of k :*

- (1) *If $k \notin (-4, 2)$ (no genuine torus intersection; the intersections degenerate at both boundary points: a tangency at $k = -4$, and at $k = 2$ the two folds coalesce at $\theta = \pi$, where $S_2(-1, y) = y^2 - 1$), the classical Deninger mechanism is expected to apply directly, and indeed $m(S_k) = r_k |L'(E_k, 0)|$ with a small rational r_k is confirmed numerically for $k = 2, 3$ and predicted then confirmed for $k = -4, -5, -6$ with $r = 2, 1, \frac{7}{2}, \frac{1}{4}, \frac{1}{8}$ (Conjecture 7.1). Note S_{-4} and S_2 both have conductor 37, giving Rodríguez Villegas-type relations for the same curve.*
- (2) *If $-4 < k < 2$ (genuine torus intersections, so m itself fails and the split integral is needed), the torsion dichotomy matches the numerical evidence exactly: a Boyd-type identity is proved for the two cases where the curve has torsion making x, y modular units — $k = 0$ ($\mathbb{Z}/5\mathbb{Z}$, $X_1(11)$) and $k = 1$ ($\mathbb{Z}/4\mathbb{Z}$) — while for $k = -1, -2, -3$ the torsion is trivial and PSLQ searches find no rational relation. (These three have branch-exchange intersections at $\theta = 0$: exactly the cases where the torus meets the curve but no modular units exist.)*

Remark 3.5. The $k = 1$ row ($\tilde{n}(1) = b_{17}$) was first found as an independent high-precision confirmation of the conductor-17 analogue stated by Samart; it is now proved as Theorem 5.1 (§5).

3.3. The conductor-53 case: negative adjudication. Samart [22]*§4 mentions a conductor-53 analogue (for $k = -1$) in a single sentence, with no formula, precision, or reference. We present three independent obstructions indicating that it cannot hold.

Proposition 3.6 (Negative adjudication of the conductor-53 case). *For S_{-1} (curve 53.a1), no Boyd-type identity of the (C3) shape — a split-integral identity on the torus proved via the anti-invariant closed cycle and the Beilinson–Brunault mechanism — can exist, and the natural numerical candidates fail:*

- (1) Topological obstruction. *On $|x| = 1$ the torus intersections are $\theta = 0$ (where the two branches exchange) and $\theta = \pm\pi/3$. The space of closed anti-invariant chains with breakpoints at the intersections and big/small branch choices is $\mathbb{Z} \cdot (1, 1, 1, 1)$: the boundary matrix of the anti-invariant symmetrization is a $\pm 1/0$ integer matrix whose kernel over $\mathbb{Q}$ is exactly one-dimensional, computed by exact rational linear algebra (`code/k53_smith.py`), and the generator $(1, 1, 1, 1)$ has period 0 — in fact exactly, not numerically: $(1, 1, 1, 1)$ is the sum of the two sheets over the full circle, and the two branches have $u = 2y + B = \pm\sqrt{B^2 - 4x^3}$, so $1/u_+ + 1/u_- \equiv 0$ pointwise and $\sum_{\text{branches}} dx/u$ vanishes identically (the continuous-root loop, which closes only after two turns, vanishes by the same cancellation). Hence there is no non-trivial anti-invariant closed cycle on the torus, so the generator of $H_1(E, \mathbb{Z})^-$ cannot be realised there and the Boyd-type torus integral object does not exist.*
- (2) Algebraic obstruction. *53.a1 has trivial torsion and rank 1, and $(0, 0)$ is a Mordell–Weil generator (`PARI ellorder` = 0): x, y are not modular units, so the Beilinson–Brunault mechanism does not apply, and no closed-cycle regulator pairing has any reason to be a rational multiple of b_{53} .*
- (3) Numerical verdict. *For the natural candidate (the half-loop L_1 , an open chain), the period ratio $0.5492906\dots$ with w_{anti} is not in the period lattice, and the regulator integral over $2\pi b_{53}$ equals $0.7392026\dots$, for which `PSLQ` searches find no rational relation.*

We conclude that Samart's conductor-53 remark is most likely a low-precision numerical false positive.

4. PROOF OF (C3)

Throughout, $E : S_0 = 0$ with invariant differential $\omega = dx/u$, $u = 2y + x^2 + 1$ (quartic model $u^2 = x^4 - 4x^3 + 2x^2 + 1$; the model constant $\kappa = 1$ is proved exactly in §6).

4.1. Modular units. The quartic model's invariants give $j = -2^{12}/11 = j(X_1(11))$. Exact group-law computation (`code/torsion.py`, exact rational arithmetic over $\mathbb{Q}$, $\mathbb{Q}(\sqrt{2})$, $\mathbb{Q}(\zeta_8)$, not numerical) on the quartic model, with its point at infinity as neutral element, for $A = (0, 0)$ on S_0 (the point $(x, u) = (0, 1)$ in quartic coordinates):

$$2A = (0, -1), \quad 4A = (1, 0) = -A \implies 5A = O.$$

The S_0 -model's projective closure has *two* points at infinity, $O_1 = [0 : 1 : 0]$ (the image of the quartic neutral element) and $Q_\infty = [1 : -1 : 0]$ (which the birational identification with $X_1(11)$ below takes to the origin O); on this model

$$\operatorname{div}(x) = [(0, 0)] + [(0, -1)] - [O_1] - [Q_\infty], \quad \operatorname{div}(y) = 3[(0, 0)] - 2[O_1] - [Q_\infty]$$

are supported on 5-torsion points ($E(\mathbb{Q})_{\text{tors}} = \mathbb{Z}/5\mathbb{Z}$, exactly the rational cusps of $X_1(11)$), so x, y are *modular units* (Manin–Drinfeld). The polynomial is tempered: its Newton face polynomials $x^3 + y$, $x^3 + x^2y$, $x^2y + y^2$, $y^2 + y$ are all cyclotomic, so $\{x, y\} \in K_2(E) \otimes \mathbb{Q}$.

4.2. The open signed chain and the split integral. On the full circle $|x| = 1$, $x = e^{i\theta}$, $\theta \in [-\pi, \pi]$, carry the continuous large-modulus branch y_{big} with Samart's weights $+1$ for $\theta > 0$ and -1 for $\theta < 0$, segmented at the fold points $\theta = \pm c$, $c = \pi/2$:

$$\tilde{\gamma} = +[y_{\text{big}} \text{ on } [-c, c]] - [y_{\text{big}} \text{ on the two outer arcs}].$$

The corrected Mahler measure along $\tilde{\gamma}$ satisfies exact identities. On the torus $\eta(x, y) = -\log |y| d\theta$, and since $y_{\text{big}} y_{\text{small}} = x^3$ has modulus 1, $\log |y_{\text{big}}| = -\log |y_{\text{small}}|$ pointwise; comparing the signed weights of $\tilde{\gamma}$ with the definition of I_{split} gives

$$\tilde{n} = -I_{\text{split}}, \quad \int_{\tilde{\gamma}} \eta(x, y) = 2\pi I_{\text{split}}$$

(numerically confirmed to 44 digits, `code/closedness_check.py`), so that

$$(C3) \iff \int_{\tilde{\gamma}} \eta(x, y) = 2\pi b_{11}.$$

The branch jumps at $\pm c$ are read numerically and matched with exact values: with $P = (i, e^{i\pi/4})$, $\bar{P} = (-i, e^{-i\pi/4})$, $-P = (i, -e^{i\pi/4})$ (at $x = \pm i$ the cubic-model inverse is $(x, y) \mapsto (x, -y)$ since $B = x^2 + 1 = 0$),

$$\partial\tilde{\gamma} = [P] - [\bar{P}] + [-P] - [-\bar{P}] =: D, \quad \deg D = 0, \quad c(D) = -D.$$

The point P is *not* torsion (proved rigorously by reduction at good primes, §6)—but closedness does not require torsion endpoints.

4.3. The closed chain lemma. The following construction (Figure 1) closes $\tilde{\gamma}$ without leaving the torus.

Lemma 4.1 (Closed chain lemma). *Let α_2 be the small-branch inner arc $\theta : c \rightarrow -c$ (endpoints $P \rightarrow \bar{P}$) and α_1 the small-branch outer arc $\theta : c \rightarrow \pi$ and $-\pi \rightarrow -c$ (endpoints $-P \rightarrow -\bar{P}$; the inner interval cannot join these two points since the small branch does not pass through them). Then for $\beta_0 = \alpha_1 + \alpha_2$,*

$$\partial\beta_0 = -D, \quad c(\beta_0) = -\beta_0,$$

hence

$$C' := \tilde{\gamma} + \beta_0 : \quad \partial C' = 0, \quad c(C') = -C'$$

is a closed, anti-invariant, integral 1-cycle.

Proof. At $x = \pm i$ the coefficient $B(x) = x^2 + 1$ vanishes and S_0 reduces to $y^2 = -x^3 = \mp i$; hence the two branch values are $y = \pm e^{\pm i\pi/4}$, i.e. the four points $P, \bar{P}, -P, -\bar{P}$. Reading the continuous branches on the torus (each assignment certified in interval arithmetic at two scales $\varepsilon = 10^{-6}, 10^{-9}$ by `code/branch_certify.py`, §6, and matched with these exact values),

$$y_{\text{big}}(c^-) = -P, \quad y_{\text{big}}(c^+) = P, \quad y_{\text{big}}(-c^+) = -\bar{P}, \quad y_{\text{big}}(-c^-) = \bar{P},$$

so the signed chain $\tilde{\gamma} = +[y_{\text{big}} \text{ on } [-c, c]] - [y_{\text{big}} \text{ on the outer arcs}]$ has boundary $\partial\tilde{\gamma} = [P] - [\bar{P}] + [-P] - [-\bar{P}] = D$. For the small branch,

$$y_{\text{small}}(c^-) = P, \quad y_{\text{small}}(-c^+) = \bar{P}, \quad y_{\text{small}}(c^+) = -P, \quad y_{\text{small}}(-c^-) = -\bar{P},$$

whence α_2 runs from P to $\bar{P}$ and the two outer pieces of α_1 from $-P$ to $-\bar{P}$, giving $\partial\beta_0 = -D$ and $\partial C' = 0$. Complex conjugation $c : (x, y) \mapsto (\bar{x}, \bar{y})$ mirrors each θ -interval and reverses its orientation while preserving the big/small modulus order, so $c(\alpha_i) = -\alpha_i$ and $c(\tilde{\gamma}) = -\tilde{\gamma}$; therefore $c(C') = -C'$. $\square$

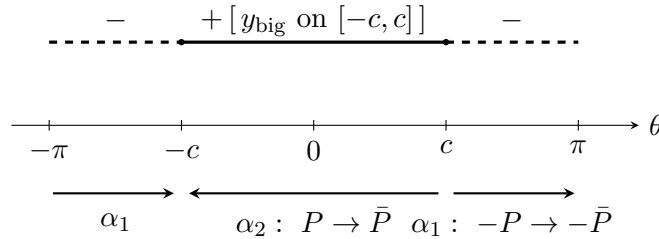


FIGURE 1. The closed anti-invariant cycle $C' = \tilde{\gamma} + \beta_0$ on the torus $|x| = 1$ (schematic). Top: Samart's signed open chain $\tilde{\gamma}$ on the big branch (weight + on $[-c, c]$, weight - on the outer arcs), with branch jumps at the fold points $\theta = \pm c$. Bottom: the compensating small-branch arcs $\beta_0 = \alpha_1 + \alpha_2$; their boundary is the negative of $\partial\tilde{\gamma}$, so C' is closed, and complex conjugation shows $c(C') = -C'$.

4.4. Homology class: $C' = 2\gamma^-$ (**certified**). In genus 1 with $\Delta < 0$, $H_1(E, \mathbb{Z})^- = \mathbb{Z}\gamma^-$, the pairing with ω is injective (period lattice isomorphism), and the period of the anti-invariant generator is the (unique up to sign) purely imaginary period of E . We fix the sign convention

$$w_{\text{anti}} := -2.9176332338769904586617792258073505143184868579 \dots i \quad (\text{PARI 11.a3}),$$

and choose γ^- with $\text{period}(\gamma^-) = w_{\text{anti}}$. Therefore $\text{period}(C')/w_{\text{anti}}$ is *a priori* an integer.

The computation (`code/n1_certify.py`, `mpmath` at 60 and 80 digits, cross-checked against PARI 11.a3 E.omega) gives

$$(2) \quad \text{period}(C') = I_{\text{signed}} + A_{s,\text{outer}} - A_{s,\text{inner}} = -5.8352664677539809 \dots i,$$

$$\frac{\text{period}(C')}{w_{\text{anti}}} = 1.999999999999999 \dots, \quad |\text{period}(C') - 2w_{\text{anti}}| = 2.5 \times 10^{-16},$$

where $I_{\text{signed}} = -2.917633233876990458 \dots i = w_{\text{anti}}$ is the period of the open signed chain and $A_{s,\text{outer}}, A_{s,\text{inner}}$ are the period integrals of the compensating arcs; the 60- and 80-digit runs agree, the residual error coming from the $1/\sqrt{\theta}$ endpoint singularity at $\theta = 0$. A-priori integrality, the distance-to-nearest-integer $2.5 \times 10^{-16} \ll 1$, and the ball-arithmetic certification of §6.2 imply

$$\boxed{\text{class}(C') = 2\gamma^-}.$$

(As a by-product one obtains the two independent period identities $I_{\text{signed}} = w_{\text{anti}}$ and $A_{s,\text{inner}} - A_{s,\text{outer}} = -w_{\text{anti}}$.)

Remark 4.2. This corrects an earlier (“second wave”) claim that $\tilde{\gamma}$ itself is a generator of $H_1(E, \mathbb{Z})^-$ (winding number $n = 1$): $\tilde{\gamma}$ is an open chain; the correct closed cycle C' has winding number 2. For contrast, the naive loop integral $I_{\text{loop}} = -0.47447 \dots i$ is not even the period of any integral cycle (ratio 0.16262... to w_{anti}).

4.5. The regulator identity along the closed chain. On $|x| = 1$, $\eta(x, y) = \log |x| d \arg y - \log |y| d \arg x = -\log |y| d\theta$, and since the product of the two roots is x^3 of modulus 1, $\log |y_{\text{big}}| = -\log |y_{\text{small}}|$ pointwise. Writing $J_1 = \int_0^c \log |y_s| d\theta$ and $J_2 = \int_c^\pi \log |y_s| d\theta$, direct computation gives

$$\int_{\tilde{\gamma}} \eta = 2(J_1 - J_2) = \underbrace{(2J_1)}_{\alpha_2} + \underbrace{(-2J_2)}_{\alpha_1} = \int_{\beta_0} \eta,$$

hence the *exact* identity

$$(3) \quad \int_{C'} \eta = 2 \int_{\tilde{\gamma}} \eta.$$

Lemma 4.3. *The integral $\int_{C'} \eta(x, y)$ depends only on the class of C' in $H_1(E, \mathbb{Z})^-$.*

Proof. Since $\{x, y\} \in K_2(E) \otimes \mathbb{Q}$ (§4, temperedness), the residues of $\eta(x, y)$ at the support of $\text{div}(x), \text{div}(y)$ are torsion, so the pairing with η descends to homology. The chain C' avoids $\text{supp div}(x) \cup \text{supp div}(y)$: those points have $x \in \{0, \infty\}$ while $C' \subset \{|x| = 1\}$; and at the fold points $\log |y| = 0$, so $\eta = -\log |y| d\theta$ is continuous and integrable along C' . $\square$

4.6. Regulator constants: Bloch, Brunault, Bertin. The divisor algebra (projective computation on the minimal cubic model $y^2 + y = x^3 - x^2$, to which the data are transported by the explicit birational change of variables; note that the two points at infinity of S_0 map to *different* points, $Q_\infty \mapsto O$ and $[0 : 1 : 0] \mapsto 3A$; Abel's principal-divisor test passed) gives, with $A = (0, 0)$ the 5-torsion point and $O = [0 : 1 : 0]$,

$$\text{div}(x) = [A] + [2A] - [O] - [3A], \quad \text{div}(y) = 3[2A] - 2[3A] - [O],$$

hence the diamond product

$$(x) \diamond (y) = 6(O) - 5(A) + 5(2A), \quad D_E((x) \diamond (y)) = -5D_E(P) + 5D_E(2P).$$

The elliptic dilogarithm values (mpmath, 60 digits; lattice basis $(w_1, w_1 - w_2)$ so that $\tau = 0.5 + 0.2299i$; $z(P) = 0.6w_1$, $z(2P) = 0.2w_1$ from PARI `ellpointtoz`):

$$D_E(P) = 0.1911937370843316957549544343121738161012 \dots$$

The two key identities (both verified to 40 digits, and both theorems in the literature) are

- **Brunault's coefficient** [7]*(3.151): $D_E(P) = \frac{2\pi}{5}b_{11} = \frac{11}{10\pi}L(E, 2)$, i.e. $L(E, 2) = \frac{10\pi}{11}D_E(P)$;
- **Bertin's exotic relation** [1, 2]: $D_E(2P) = \frac{3}{2}D_E(P)$ (note $\frac{3}{2}$, not the naive 2).

The regulator of $\{x, y\}$ along γ^- is now evaluated *directly* by a theorem of Bertin and Brunault, without any normalization choice on our part. One identification must be made explicit: the symbol of Brunault's (3.210) is *not* ours. His $\{x, y\}$ is the pair of Weierstrass coordinate functions of $E : y^2 + y = x^3 - x^2$ [7]*Thm. 8, (3.211); on this model

$$\text{div}(x_W) = [A] + [4A] - 2[O], \quad \text{div}(y_W) = 2[A] + [3A] - 3[O],$$

hence, in $\mathbb{Z}[E]^-$,

$$(x_W) \diamond (y_W) \equiv 8(O) + 5(A) - 5(2A), \quad D_E((x_W) \diamond (y_W)) = -\frac{5}{2}D_E(P)$$

by Bertin's exotic relation. Comparing with the expansion above,

$$(4) \quad D_E((x_W) \diamond (y_W)) = -D_E((x) \diamond (y)) \neq 0,$$

the non-vanishing since $D_E(P) = \frac{2\pi}{5}b_{11} \neq 0$. Both symbols are tempered: their tame symbols at the divisor supports are roots of unity (exact local-expansion computation: $T \in \{\pm 1\}$ at $A, 2A, 3A, O$ for $\{x, y\}$; $T_A = -1$, $T_O = +1$ for $\{x_W, y_W\}$; `code/bertin_diamond.py`), so Bloch's theorem

applies to both. In *any* of its normalizations Bloch's theorem evaluates the regulator of a tempered symbol as D_E of its diamond product times a constant that depends only on the fixed pair (E, γ^-) and the conventions — not on the symbol — so the ratio of the two evaluations is forced:

$$\frac{\int_{\gamma^-} \eta(x, y)}{\int_{\gamma^-} \eta(x_W, y_W)} = \frac{D_E((x) \diamond (y))}{D_E((x_W) \diamond (y_W))} = -1.$$

With Brunault's normalization [7]*(3.211)

$$r_\gamma\{f, g\} := \frac{1}{2\pi} \int_\gamma \eta(f, g), \quad \gamma \text{ generating } H_1(E, \mathbb{Z})^-,$$

Brunault's (3.210), quoting Bertin's [2]*Th. 6 and Cor. 6.1, gives

$$|r_{\gamma^-}\{x_W, y_W\}| = \frac{5}{2\pi} D_E(P) = b_{11}, \quad \text{i.e.} \quad \int_{\gamma^-} \eta(x_W, y_W) = \pm 2\pi b_{11},$$

the sign depending on orientation (Brunault, Remarque 20: D_E depends on the orientation of $E(\mathbb{R})$ and is defined only up to sign). We certified this evaluation independently: direct numerical integration of $\eta(x_W, y_W)$ along an anti-invariant generator $\gamma^- = s(2w_2 - w_1)$, $0 \leq s \leq 1$, gives $-2\pi b_{11}$ to 9 Richardson-extrapolated digits (`code/bertin_diamond.gp`). By the ratio,

$$(5) \quad \int_{\gamma^-} \eta(x, y) = \pm 2\pi b_{11}.$$

The regulator side is therefore a proved theorem; the gap closed here is the identification of Boyd's split-integral chain with the closed cycle in $H_1(E, \mathbb{Z})^-$.

Remark 4.4 (The factor 2 in the $\diamond$ -form of Bloch's theorem). The formal expansion above evaluates to $D_E((x) \diamond (y)) = \frac{5}{2} D_E(P) = \pi b_{11}$, while the certified integral is $\int_{\gamma^-} \eta = 2\pi b_{11}$: in the factor-1 normalization $r(\{x, y\})[\gamma] = D^E((x) \diamond (y))$, $r[\gamma] = \int_\gamma \eta$, of [12]*Thm. 6 the two sides differ by exactly 2. We investigated this discrepancy with three *independent* tempered symbols on the same curve (`code/bertin_diamond.py`, `code/bertin_diamond.gp`, archive `notes/attack13-bertin-diamond.txt`): the Weierstrass symbol above, our S_0 -symbol, and the coordinate symbol $\{X, Y\}$ of Bertin's cubic $(X+1)(Y+1)(X+Y+1) + XY = 0$, which maps to E by an explicit Riemann–Roch transformation (PARI `ellidentify` certifies the image model as 11.a3). The $\diamond$ -values are $-\frac{5}{2}$, $+\frac{5}{2}$ and $\frac{35}{2}$ times $D_E(P)$ respectively, while direct integration of η along γ^- gives -5 , $+5$ and $+35$ times $D_E(P)$ (the first two certified as above; the third by extrapolated Riemann sums, consistent with $2\pi m(C_1) = 14\pi b_{11}$). The factor is exactly 2 in all three cases: it is a property of (E, γ^-) , not of the symbol, the $\diamond$ -convention (identical on both curves), the tame symbols (all roots of unity), or the D_E series. The conductor-17 computation of §5, run with the same conventions

and the same code, closes with factor 1 — corroborated there by the proved theorem of [12]. The two curves differ precisely in their real topology: E has $\Delta = -11 < 0$, a single real component and $\Re\tau = 1/2$ (so $\gamma^- = a - 2b$ rather than the real locus), while the conductor-17 curve has $\Delta > 0$, two real components and $\Re\tau = 0$; we expect the factor to originate in the normalization of anti-invariant cycles against the fundamental-domain boundary in Bloch's proof for one-component lattices, but have not pursued a mechanism-level proof. Since the proof of (C3) uses only the ratio (4), in which any such curve-level constant cancels, this question does not affect the main theorem; we record it explicitly. A by-product: $D_E((X) \diamond (Y)) = 7 D_E((x) \diamond (y))$ exactly, which explains the coefficient 7 in Bertin's $m(C_1) = 7b_{11}$ at the level of $K_2(E)$. Finally, the form $\pi r = D_E(\cdot)$ found in expository accounts (e.g. [24]*Thm. 1) is consistent with neither the factor-1 nor the factor-2 data above and is not used anywhere in this paper.

4.7. Synthesis.

Theorem 4.5 ((C3)). $I_{\text{split}} = b_{11}$.

Proof. By (3) and $\text{class}(C') = 2\gamma^-$ (§4.4), and the anchor (5),

$$\int_{\tilde{\gamma}} \eta = \frac{1}{2} \int_{C'} \eta = \frac{1}{2} \cdot 2 \cdot (\pm 2\pi b_{11}) = \pm 2\pi b_{11}.$$

The sign is pinned by a single numerical evaluation: $\int_{\tilde{\gamma}} \eta = +0.9559686854\dots = +2\pi b_{11}$ with $b_{11} > 0$. Finally, by the identity $I_{\text{split}} = \frac{1}{2\pi} \int_{\tilde{\gamma}} \eta$ of §4,

$$I_{\text{split}} = b_{11}. \quad \square$$

5. THE CONDUCTOR-17 CASE: $k = 1$

The proof of (C3) in §4 is a *method*, and its first re-application is Samart's conductor-17 analogue. For $k = 1$ the polynomial

$$S_1 = y^2 + (x^2 + x + 1)y + x^3$$

defines an elliptic curve of conductor 17 (PARI `ellfromeqn` gives $E_1 : y^2 + xy - y = x^3 - x^2$, $\Delta = 17$; `ellglobalred` confirms the conductor), with $E_1(\mathbb{Q})_{\text{tors}} = \mathbb{Z}/4\mathbb{Z}$ generated by $A = (0, 0)$, which has order 4 — the exact parallel of the 5-torsion point of the $k = 0$ case. Every step of §4 goes through, and the regulator side is in fact *easier*: the key D_E -value is a published theorem of Lalín–Ramamonjisoa [12].

5.1. The closed cycle. The torus-intersection (fold) angle is $c = \arccos(-k/2)|_{k=1} = 2\pi/3$. At $\theta = c$, $x = \omega = e^{2\pi i/3}$ satisfies $x^2 + x + 1 = 0$, so S_1 reduces to $y^2 + 1 = 0$: the branch jump values are *exactly* $y = \pm i$ (the $k = 0$ analogue was $y^2 = i$). Writing $P = (\omega, i)$, the signed chain $\tilde{\gamma}$ (big branch on $[-c, c]$ minus the big branch on the two outer arcs) has boundary

$$\partial\tilde{\gamma} = [P] - [\bar{P}] + [-P] - [-\bar{P}],$$

and the same small-branch compensating arc β_0 closes it to a cycle $C' = \tilde{\gamma} + \beta_0$ with $\partial C' = 0$, $c(C') = -C'$. The eight endpoint branch assignments and the modulus ordering are certified in interval arithmetic exactly as for $k = 0$ (`code/k1_branch_certify.py`, output `notes/attack12-k1-branch.txt`: all 8 assignments certified at $\varepsilon = 10^{-6}, 10^{-9}$; 113 strict bisection balls plus 2 special balls at the corner $\theta = c$).

One structural difference from $k = 0$: here $\Delta = +17 > 0$, so $E(\mathbb{R})$ has two connected components, $\overline{w_2} = -w_2$ holds on the nose, and the primitive anti-invariant period is simply

$$w_{\text{anti}} = w_2 = -2.7457391180897536720341879 \dots i$$

(for $k = 0$, $\Delta < 0$ forced $w_{\text{anti}} = 2w_2 - w_1$). The model constant is trivial here: the `ellfromeqn` model already has $\Delta = 17 = \Delta_{\min}$, so the transformation to the minimal model has $u = \pm 1$, the period lattices coincide ($\kappa = \pm 1$; the sign only flips the displayed ratio, not its integrality — cf. the $k = 0$ discussion in §6).

5.2. Homology class: interval-arithmetic proof. As in §4.4, $\text{period}(C')/w_{\text{anti}}$ is a-priori an integer. The Arb certification (script `code/k1_interval.py`; output `attack11-k1-interval.txt`, 300-bit working precision) is simpler than for $k = 0$ because $D(z) = z^4 - 2z^3 + 3z^2 + 2z + 1$ has *no zeros on* $|z| = 1$ ($\min_{|z|=1} |D| = 4 > 0$, certified): there is no endpoint singularity and no tip-estimate machinery. The period w_{anti} is certified independently of PARI via Newton–Rouché isolation of the roots of $4x^3 - 3x^2 - 2x + 1 = (x - 1)(4x^2 + x - 1)$ and the two-component Carlson R_F formulas. The resulting ratio ball is

$$[-2.00 \pm 2.4 \times 10^{-14}] + [\pm 2.4 \times 10^{-14}] i, \quad |\text{ratio} + 2| \leq 3.4 \times 10^{-14} < \frac{1}{2},$$

(the radius is set by the integration tolerance, not by the arithmetic). Choosing γ^- with $\text{period}(\gamma^-) = w_{\text{anti}}$, the a-priori integrality upgrades to the exact equality

$$\text{class}(C') = 2\gamma^- \quad (\text{the script uses } -w_{\text{anti}}, \text{ displaying } -2).$$

5.3. The regulator side. The integral algebra of §4 is unchanged: the pointwise identity $\log |y_{\text{big}}| = -\log |y_{\text{small}}|$ on $|x| = 1$ gives $\int_{\beta_0} \eta = \int_{\tilde{\gamma}} \eta$, hence $\int_{C'} \eta = 2 \int_{\tilde{\gamma}} \eta$. The divisors have the same shape

$$\text{div}(x) = [A] + [2A] - [O] - [3A], \quad \text{div}(y) = 3[A] - 2[O] - [3A],$$

now verified *four* ways: local expansions, PARI, code expansion, and an explicit birational map $X = -(x + y)$, $Y = x(x + y)$ from S_1 to E_1 : $Y^2 + XY - Y = X^3 - X^2$. The diamond product (definition as in [13]) expands exactly to

$$(x) \diamond (y) \equiv 6(O) + 4(A) - 6(2A) \quad \text{in } \mathbb{Z}[E]^+.$$

The two D_E -values are settled:

- $D_E(2A) = 0$ *rigorously*: $2A$ is 2-torsion, q is a positive real, and the Bloch–Wigner series vanishes termwise at $z_q = -1$;
- $D_E(A) = \frac{17}{8\pi}L(E_1, 2)$ is a *published theorem*: Lalín–Ramamonjisoa [12]*§5 prove $L(E_{17}, 2) = \frac{8\pi}{17}D^E(P)$ for the 4-torsion generator P of their model, in the D^E -normalisation (their Def. 5, eq. (10)) that coincides verbatim with our series implementation; we reproduced the value independently to 60 digits.

Hence

$$D_E((x) \diamond (y)) = 4D_E(A) = \frac{17}{2\pi}L(E_1, 2) = 2\pi b_{17},$$

$$b_{17} = L'(E_1, 0) = \frac{17}{4\pi^2}L(E_1, 2).$$

5.4. Synthesis. We use Bloch's theorem in the normalisation of [12]*Thm. 6: for $\{x, y\} \in K_2(E) \otimes \mathbb{Q}$ and γ^- generating $H_1(E, \mathbb{Z})^-$,

$$\int_{\gamma^-} \eta(x, y) = \pm D^E((x) \diamond (y)).$$

The temperedness hypothesis holds: the four face polynomials of S_1 are $x^3 + y$, $x^3 + x^2y$, $x^2y + y^2$ and $y^2 + y$, all cyclotomic. This normalisation has published precedent on this very curve: the conductor-17 identity of [12] combined with Zudilin's $m = 2b_{17}$ closes with the same factor.

Theorem 5.1 (Samart's conductor-17 analogue). $\tilde{n}(1) = b_{17}$.

Proof. By the above,

$$\int_{\tilde{\gamma}} \eta = \frac{1}{2} \int_{C'} \eta = \frac{1}{2} \cdot 2 \cdot (\pm D_E((x) \diamond (y))) = \pm 2\pi b_{17},$$

and the sign is pinned by a single numerical evaluation:

$$\int_{\tilde{\gamma}} \eta = -1.8809066251248561 \dots = -2\pi b_{17}.$$

The structural identity $\tilde{n}(1) = -\frac{1}{2\pi} \int_{\tilde{\gamma}} \eta$ (the $k = 1$ instance of the pointwise argument of §4) gives $\tilde{n}(1) = b_{17}$. $\square$

Remark 5.2 (The normalization of Bloch's theorem). For the conductor-17 proof, Bloch's theorem is used in the normalization of [12]*Thm. 6: $r(\{x, y\})[\gamma] = D^E((x) \diamond (y))$ with $r[\gamma] = \int_{\gamma} \eta$, γ a generator of $H_1(E, \mathbb{Z})^-$. That this (and not a factor-2 variant) is the correct statement *on that curve* follows from a consistency argument in the conductor-17 case (our own reconstruction; in [12] the constant C below is never determined): in [12]*§7 the class of the real cycle is an *a priori* unknown integer multiple C of γ , and comparing the factor-1 theorem applied to their $(X) \diamond (Y)$ with their proved Corollary 2 (their Thm. 1, eq. (5), combined with Zudilin's identity, their eq. (6)) forces $C \cdot f = 1$ with $C \in \mathbb{Z} \setminus \{0\}$, hence $f = 1$; a factor-2 statement would force $C = \frac{1}{2} \notin \mathbb{Z}$. The underlying D_E -identity of their

§5 we reproduced to 60 digits. For $k = 0$ the $\diamond$ -form enters only through the ratio (4) of two tempered symbols on the same curve, in which the normalization constant — whatever its value — cancels; the absolute value is anchored by the Bertin–Brunault theorem $|r_{\gamma^-}\{x_W, y_W\}| = \frac{5}{2\pi}D_E(P)$ with $r_{\gamma^-} = \frac{1}{2\pi} \int_{\gamma^-} \eta$ (Brunault (3.210)–(3.211), §4). Remarkably, the factor-1 statement itself *fails* on the conductor-11 curve: all three independent tempered symbols tested there satisfy $\int_{\gamma^-} \eta = 2 D_E((\cdot) \diamond (\cdot))$, a curve-level phenomenon discussed in Remark 4.4; this does not affect either proof, since the conductor-17 argument is self-contained on its own curve and the conductor-11 argument uses only the ratio. The form $\pi r = D_E(\cdot)$ found in expository accounts is consistent with neither data set and is not used anywhere in this paper.

6. NUMERICAL CERTIFICATION

6.1. Certified computations. All floating-point computations are performed with `mpmath` at 60–300 digits and cross-checked with `PARI/GP` 2.15.5:

- **The constant b_{11} (`code/b11.py`):** from the exact integer coefficients of $f_{11} = \eta(\tau)^2 \eta(11\tau)^2$ (truncated convolution of the squared Euler function) via the approximate functional equation for weight 2, root number +1:

$$b_{11} = \Lambda(f, 2) = \sum_{n \geq 1} a_n \left[e^{-t_n} \left(\frac{1}{t_n} + \frac{1}{t_n^2} \right) + E_1(t_n) \right], \quad t_n = \frac{2\pi n}{\sqrt{11}},$$

with terms decaying like $e^{-1.894n}$; the tail bound drives the truncation (442 terms at 800 digits of working precision).

- **(C3) to 366 digits (`code/attack13_c3_300.py`, `archive/notes/attack13-c3-300.txt`):**

$$I_{\text{split}} = 0.15214714172591804948622729747863449562814 \\ 3589164226122809889823882023289695302776676 \dots,$$

$$|I_{\text{split}} - b_{11}| = 9.26 \times 10^{-367},$$

with the η -product series value of b_{11} agreeing digit-for-digit with `PARI/GP` `lfun(E, 0, 1)` at 330 digits.

- **(C1)/(C2) reproduced to 52 digits (`code/attack1.py`):** $\|m - 7b_{11}\| \approx 5.0 \times 10^{-53}$, $\|m - 5b_{11}\| \approx 3.6 \times 10^{-53}$.
- **Period certification (`code/n1_certify.py`, `code/n1_certify.gp`):** (2) computed independently at 60 and 80 digits with identical results; each arc integral A_s is stable to 24 digits across 40/60/80-digit runs; w_{anti} taken from `PARI E.omega` for `11.a3`. The integrality argument has an a-priori gap of 1 against a residual of 2.5×10^{-16} .
- **Exact algebraic checks:** group law and torsion (`code/torsion.py`) use exact rational arithmetic over $\mathbb{Q}$, $\mathbb{Q}(\sqrt{2})$, $\mathbb{Q}(\zeta_8)$ —no numerical approximation. Non-torsion of the endpoint P is proved *rigorously*

(`code/endpoint_torsion3.py`) by reduction at good primes: $\text{ord}(P \bmod 17) = 20$ and $\text{ord}(P \bmod 89) = 3$ with $\gcd(20, 3) = 1$, so P cannot have finite order (reduction is injective on prime-to- p torsion, Silverman VII.3.1); cross-checked by exact $\mathbb{Q}(\zeta_8)$ arithmetic and PARI `ellorder`. The model constant $\kappa = 1$ is an *exact rational identity* (`code/kappa_exact.py`): an explicit birational map to the minimal model, derived by Riemann–Roch and composed with PARI's exact minimal transform, satisfies $2Y + 1 = u dX/dx$ as a polynomial identity, hence $\omega_{\min} = dX/(2Y + 1) = dx/u$ exactly (sympy-verified, including the inverse round-trip).

- **Branch certification** (`code/branch_certify.py`, output `notes/attack12-branch.txt`): the eight endpoint branch assignments of the closed chain lemma are certified in ball arithmetic at $\varepsilon = 10^{-6}$ and 10^{-9} (certified distance $< 1/10$ to the claimed endpoint, with $D \neq 0$ certified on the closing ball so continuity pins the limit), and the modulus ordering of Proposition 3.1 is certified by adaptive bisection of $[0, \pi]$ (153 strict balls with $D \neq 0$ and $|y_{\text{small}}| < 1 < |y_{\text{big}}|$, plus the special balls at 0 and $\pi/2$ where equality holds exactly).
- **The Bertin-symbol adjudication** (`code/bertin_diamond.py`, `code/bertin_diamond.gp`, archive `notes/attack13-bertin-diamond.txt`): all divisor, diamond-product and tame-symbol computations of §4 are exact (sympy rational arithmetic, Abel's principal-divisor test passed); the Riemann–Roch map from Bertin's cubic to E is certified by PARI `ellidentify` (11.a3, exact change of variables $[1, -1, -2, 2]$); the D_E -values and the ratio -1 of (4) are evaluated at 60 digits; the two regulator integrals of Remark 4.4 are computed independently of the main pipeline.
- **PARI/GP cross-checks**: `verify_family.gp`, `verify_ratios.gp` (family conductors, j -invariants; the $\tilde{n}(k)$ and $m(S_k)$ constants for $k = 2, 3$ recomputed to 70 digits and matched against PARI `lfun` to all 70 digits), `winding.gp`, `dilog.gp`, `k53.gp`, `k53b.gp`, `kfamily_torsion.gp`.
- **Conductor-17 certification** (scripts `code/k1_*`, listed individually in Appendix A): PARI curve identification and z -values; mp-math 60/80-digit chain integrals (ratio 2.0000000000000002); exact diamond-product expansion with the $k = 0$ case as control; $D_E(A)$, $D_E(2A)$ to 60 digits; and the Arb ironclad proof of the period ratio (§5).

6.2. Level of rigour: interval certification. The integer identification in §4.4 has been made fully interval-rigorous (script `code/n1_interval.py`, python-flint/Arb ball arithmetic [10], 300-bit working precision; output in `notes/attack10-interval.txt`). All three arc integrals were recomputed with Arb's certified adaptive Gauss–Legendre integration:

- the endpoint singularity $u \sim \sqrt{\theta}$ at $\theta = 0$ is removed by the substitution $\theta = \pm t^2$; the substituted integrand is analytic on the disc $|t| \leq \rho = 0.6$ (the nonzero zeros of $D(t^2)$ satisfy $|\theta_j| \geq 1.2187$, certified by Newton–Rouché isolation of the roots of $z^4 - 4z^3 + 2z^2 + 1$), and the tip $[0, \delta]$, $\delta = \rho/5$, is summed by the Cauchy estimate $\int_0^\delta f = a_0\delta + R$, $|R| \leq H\delta^3/3$, where H is derived from $M = \max_{|t|=\rho} |f| = 1.3075\dots$, itself certified by covering the circle with 4096 balls (per-tip Cauchy remainder bound $H\delta^3/3 = 2.18 \times 10^{-3}$). The tip constant a_0 , obtained by hand from the local expansion, is not taken on faith: the script re-evaluates f by ball arithmetic at $t = \delta$ and certifies $|f(\delta) - a_0| \leq H\delta^2$ for each of the four tips (a sign or branch error in a_0 would move the value by 2, far outside this bound);
- $D(\theta)$ crosses the negative real axis once on (c, π) and once on $(-\pi, -c)$, where the principal square root is not analytic: the arcs are subdivided adaptively, on each piece either $\sqrt{D}$ or $i\sqrt{-D}$ is certified analytic by ball evaluation, and the overall sign is propagated across the nodes by a certified matching test;
- w_{anti} is certified independently of PARI: the roots of $4x^3 - 4x^2 + 1$ are isolated by Newton iteration plus a Rouché certificate, and Carlson’s R_F (DLMF §19) gives $w_{\text{real}} = 6.34604652139776710844397\dots$ and the purely imaginary period $2.91763323387699045866178\dots i$ — which is $-w_{\text{anti}}$ in the convention of §4.4 — with certified radii 6.7×10^{-48} and 4.9×10^{-49} , agreeing with PARI to 45 digits.

Remark 6.1 (Two implementation disclosures). (i) The ball radii displayed in the archived outputs (e.g. `notes/attack10-interval.txt`) follow the Arb/python-flint `repr` convention, which folds midpoint printing error into the displayed radius; the true ball radii are smaller (the displayed interval contains the true ball). All certified bounds quoted above use the exact radii, not the displayed ones. (ii) An earlier version of the arc-integral code could in principle take the wrong square-root sheet on the first sub-arc when the branch-cut avoidance selects the $i\sqrt{-D}$ variant (this latent issue was found and fixed during the $k = 1$ work; the $k = 0$ case never triggers it because its cut crossings lie in the interior of sub-arcs, whose first piece always uses the $\sqrt{D}$ variant). The $k = 0$ script was nevertheless re-audited against the corrected logic; had the issue been triggered it would flip the sign of an entire arc, moving the ratio off -2 by an $O(1)$ amount — the archived output shows this did not occur.

Against w_{anti} the resulting ratio ball is $[2.00 \pm 3.13 \times 10^{-3}] + [\pm 2.99 \times 10^{-3}] i$: it contains 2 and $|\text{ratio} - 2| \leq 4.33 \times 10^{-3} < 1/2$. The a-priori integrality of §4.4 therefore upgrades to the exact equality $\text{period}(C') = 2w_{\text{anti}}$, i.e. $\text{class}(C') = 2\gamma^-$. (The script itself works with the opposite sign of the period throughout, so the raw output in `notes/attack10-interval.txt` displays the ratio ball centred at -2 ; the two formulations are equivalent.)

7. CONJECTURES

The family identities established numerically here are stated as conjectures pending analytic proof:

Conjecture 7.1. *For $S_k = y^2 + (x^2 + kx + 1)y + x^3$ with $k \notin (-4, 2)$ (integer k), there is a small rational number r_k such that*

$$m(S_k) = r_k |L'(E_k, 0)|.$$

Confirmed numerically: $r_2 = 2$, $r_3 = 1$ (70 digits), and $r_{-4} = \frac{7}{2}$ (conductor 37), $r_{-5} = \frac{1}{4}$ (conductor 359), $r_{-6} = \frac{1}{8}$ (conductor 997) (25 digits each).

The last three deserve emphasis as *predicted then confirmed*: the structural dichotomy (Proposition 3.4) predicted, from the torus-intersection trichotomy, that $k = -4, -5, -6$ must satisfy Boyd-type identities with specific small rationals, and the subsequent computation hit the predicted values. Note that S_{-4} and S_2 share conductor 37, yielding Rodríguez Villegas-type rational relations [20, 12] between two different polynomial models of the same curve.

Remark 7.2 (Samart's conductor-17 analogue: now a theorem). The conjecture $\tilde{n}(1) = b_{17}$ for $k = 1$ (conductor 17), suggested by Samart and confirmed numerically to 60 digits, is proved as Theorem 5.1 in §5.

Remark 7.3. By Theorem 3.6, no such conjecture can be formulated for $k = -1$ (conductor 53): the anti-invariant cycle on the torus does not exist and x, y are not modular units.

APPENDIX A. REPRODUCTION CODE

All scripts live in `code/`; raw outputs in `notes/attack*.txt`. Dependencies: Python 3.12 with mpmath and sympy; PARI/GP 2.15.5; python-flint 0.9.0 (Arb) for the interval certification.

File	Purpose
b11.py	$b_{11} = L'(E_{11}, 0)$, 300 digits (functional equation)
attack1.py	reproduce (C1), (C2) to 52 digits
attack2.py	$m(S_0)$, PSLQ negative results
attack3.py	(C3) to 149 digits (superseded)
attack13_c3_300.py	(C3) to 366 digits, PARI lfun cross-check
bertin.diamond.py/.gp	Bertin symbol: exact $\diamond$, tame, D_E -ratio -1
torsion.py	exact group law: $5A = O$, modular units
endpoint_torsion3.py	rigorous non-torsion of P (reduction mod p)
kappa_exact.py	exact proof $\kappa = 1$ (birational map, $2Y + 1 = u dX/dx$)
boundary_torsion.py	boundary-divisor torsion analysis
closedness.check.py	$\tilde{n} = -I_{\text{split}}$, regulator $= 2\pi I_{\text{split}}$
ntilde.family.py	$\tilde{n}(k)$ family table, 50–80 digits
b_family.py	b_N for the family via point counting
winding.py / winding.gp	period pairings, winding-number heuristics
dilog.py / dilog.gp	elliptic dilogarithm $D_E(P)$, Brunault constants
k53_attack.py	conductor-53 obstruction analysis
k53.gp / k53b.gp	PARI checks for 53.a1
kneg_m.py	$m(S_k)$ for $k = -4, -5, -6$ (predicted identities)
n1.certify.py / n1_certify.gp	certification of $\text{class}(C') = 2\gamma^-$
n1.interval.py	Arb ironclad proof of the period ratio
branch.certify.py	branch assignments + modulus ordering (certified)
k1_pari/points/zvals.gp	conductor-17 identification, torsion, z -values
k1.certify.py	mpmath certification of the $k = 1$ chain integrals
k1.branch.certify.py	$k = 1$ branch assignments + modulus ordering (certified)
k1.interval.py	Arb ironclad proof of the $k = 1$ period ratio
k1.diamond.py	exact diamond product for $k = 1$ ($k = 0$ control)
k1.dilog.py	$D_E(A)$, $D_E(2A)$ for conductor 17
verify_family.gp	family conductors and j -invariants
verify_ratios.gp	PARI lfun cross-check of ratios
kfamily_torsion.gp	torsion, $\text{ord}(0, 0)$ for the family

Reproduction:

```
cd code && python b11.py && python attack1.py && python attack2.py \
  && python attack3.py && python torsion.py && python endpoint_torsion3.py \
  && python kappa_exact.py \
  && python boundary_torsion.py && python closedness_check.py \
  && python ntilde_family.py && python b_family.py && python winding.py \
  && python dilog.py && python k53_attack.py && python kneg_m.py \
  && python n1_certify.py
.venv/Scripts/python n1_interval.py    # Arb certification, k=0
.venv/Scripts/python branch_certify.py # branch assignments + mod ordering
.venv/Scripts/python k1_interval.py    # Arb certification, conductor 17
.venv/Scripts/python k1_branch_certify.py # k=1 branch assignments + ordering
python k1_certify.py && python k1_diamond.py && python k1_dilog.py
gp -q verify_family.gp && gp -q verify_ratios.gp
gp -q winding.gp && gp -q dilog.gp
gp -q k53.gp && gp -q k53b.gp && gp -q kfamily_torsion.gp
gp -q k1_pari.gp && gp -q k1_points.gp && gp -q k1_zvals.gp
```

REFERENCES

- [1] M.-J. Bertin, *Mesure de Mahler et régulateur elliptique: preuve de deux relations "exotiques"*, in: Number theory, CRM Proc. Lecture Notes **36**, Amer. Math. Soc., Providence, RI, 2004, pp. 1–12.
- [2] M.-J. Bertin, *Mesure de Mahler d'une famille de polynômes*, J. Reine Angew. Math. **569** (2004), 175–188.
- [3] M.-J. Bertin and M. Lalin, *Mahler measure of multivariable polynomials*, in: Women in Numbers 2: Research Directions in Number Theory, Contemp. Math. **606**, Amer. Math. Soc., Providence, RI, 2013, pp. 125–147.
- [4] S. J. Bloch, *Higher regulators, algebraic K-theory, and zeta functions of elliptic curves*, CRM Monograph Series **11**, Amer. Math. Soc., Providence, RI, 2000.
- [5] D. W. Boyd, *Mahler's measure and special values of L-functions*, Experiment. Math. **7** (1998), no. 1, 37–82.
- [6] D. W. Boyd, *Mahler measure and L-functions*, slides of a talk at the Pacific Northwest Number Theory Conference, Eugene, May 15, 2015.
- [7] F. Brunault, *Étude de la valeur en $s = 2$ de la fonction L d'une courbe elliptique*, thèse, Université Paris 7 – Denis Diderot, 2005.
- [8] F. Brunault, *Valeur en 2 de fonctions L de formes modulaires de poids 2: théorème de Beilinson explicite*, Bull. Soc. Math. France **135** (2007), no. 2, 215–246.
- [9] C. Deninger, *Deligne periods of mixed motives, K-theory and the entropy of certain $\mathbb{Z}^n$ -actions*, J. Amer. Math. Soc. **10** (1997), no. 2, 259–281.
- [10] F. Johansson, *Arb: efficient arbitrary-precision midpoint-radius interval arithmetic*, IEEE Trans. Comput. **66** (2017), no. 8, 1281–1292.
- [11] F. Johansson et al., *mpmath: a Python library for arbitrary-precision floating-point arithmetic*, <http://mpmath.org/>.
- [12] M. Lalin and F. Ramamonjisoa, *The Mahler measure of a Weierstrass form*, Int. J. Number Theory **13** (2017), no. 8, 2195–2214.
- [13] M. Lalin, D. Samart and W. Zudilin, *Further explorations of Boyd's conjectures and a conductor 21 elliptic curve*, J. Lond. Math. Soc. (2) **93** (2016), no. 2, 341–360.
- [14] M. N. Lalin and M. D. Rogers, *Functional equations for Mahler measures of genus-one curves*, Algebra Number Theory **1** (2007), no. 1, 87–117.
- [15] A. Mellit, *Elliptic dilogarithms and parallel lines*, J. Number Theory **204** (2019), 1–24.
- [16] M. Rogers, *Hypergeometric formulas for lattice sums and Mahler measures*, Int. Math. Res. Not. IMRN (2011), no. 17, 4027–4058.
- [17] M. Rogers and W. Zudilin, *On the Mahler measure of $1 + X + 1/X + Y + 1/Y$* , Int. Math. Res. Not. IMRN (2014), no. 9, 2305–2326.
- [18] The LMFDB Collaboration, *The L-functions and Modular Forms Database*, <https://www.lmfdb.org>, 2026.
- [19] The PARI Group, *PARI/GP version 2.15.5*, Univ. Bordeaux, 2024, <http://pari.math.u-bordeaux.fr/>.
- [20] F. Rodríguez Villegas, *Modular Mahler measures. I*, in: Topics in number theory (University Park, PA, 1997), Math. Appl. **467**, Kluwer Acad. Publ., Dordrecht, 1999, pp. 17–48.
- [21] M. Rogers and W. Zudilin, *From L-series of elliptic curves to Mahler measures*, Compos. Math. **148** (2012), no. 2, 385–414.
- [22] D. Samart, *Mahler measure of a non-reciprocal family of elliptic curves*, Q. J. Math. **74** (2023), no. 4, 1187–1208.
- [23] C. J. Smyth, *On measures of polynomials in several variables*, Bull. Austral. Math. Soc. **23** (1981), no. 1, 49–63.
- [24] N. Touafek, *From the elliptic regulator to exotic relations*, An. Ştiinţ. Univ. "Ovidius" Constanţa Ser. Mat. **16** (2008), no. 2, 117–126.

- [25] W. Zudilin, *Regulator of modular units and Mahler measures*, Math. Proc. Cambridge Philos. Soc. **156** (2014), no. 2, 313–326.

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